

Protection of Prenylated Flavonoids from Mori Cortex Radicis (Moraceae) against Nitric Oxide-induced Cell Death in Neuroblastoma SH-SY5Y Cells

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Seven prenylated flavanoids, licoflavone C (1), cyclomulberrin (2), neocyclomorusin (3), sanggenon I (4), morusin (5), kuwanon U (6) and kuwanon E (7), and three 2-arylbenzofurans, moracin P (8), moracin O (9), and mulberrofuran Q (10) were isolated from the MeOH extract of Mori Cortex Radicis. Among these, compounds 2-7 enhanced cell viability in a dose-dependent manner against sodium nitroprusside-induced cell death in neuroblastoma SH-SY5Y cells, which was measured by MTT reduction assay (EC₅₀ values of 4.4, 5.6, 8.0, 6.4, 8.7, and 11.9 µg/mL, respectively). Among 10 compounds, C-3 prenylated flavones (2, 3, and 5) and prenylated flavanones (4, 6, and 7) showed cell protection. However, compound 1 which lacks the prenyl group at C-3 and three 2-arylbenzofurans (8-10) did not show protective effect. The order of cell protection was as follow: C-3 prenylated flavones (2, 3, and 5) > prenylated flavanones (4, 6, and 7) > 2-arylbenzofurans (8-10) and flavone (1). From this result, we show that some prenylated flavones and flavanones might protect neuronal cells against nitrosative stress-mediated cell death. Even though further evaluations are necessary *in vitro* and *in vivo* study, we carefully suggest that some prenylated flavonoids from Mori Cortex Radicis might protect neuronal cells from neurodegenerative diseases.

Key words: Mori cortex radicis, Prenylated flavonoids, Nitric oxide, Neurodegenerative disease

INTRODUCTION

The Mori Cortex Radicis (Moraceae) has long been used in oriental medicine as an antidiabetic, a diuretic, an expectorant and a laxative agent (Yadav and Nade, 2008). It has been mainly used for the treatment of constipation, cough, dizziness and urinary incontinence. And it has been used to treat fever, protect the

liver, improve eyesight, strengthen joints, discharge of urine and lower blood pressure and protect from neuronal damages (Katsube et al., 2010). A number of prenylated flavonoids, stilbenes, benzofurans, terpenes, alkaloids, and other phenolic compounds were previously reported as chemical constituents of this plant (Nomura et al., 1982; Ueda et al., 1982, 1984; Hano and Nomura, 1986; Hano et al., 1986, 1989a, 1989b; Qiu et al., 1996; Asano et al., 2001; Kusano et al., 2002; Lee et al., 2002; Zhang et al., 2009). And various prenylated flavonoids, benzofurans and other phenolic compounds isolated from these species have biological activities including cytotoxicity, inhibition of COX-1 and 2, and NO production (Nomura et al., 1983; Shi et al., 2001; Su et al., 2002; Sohn et al., 2004).

Nitric oxide (NO•) is crucial reactive radical that acts as an oxidative biological signaling molecule in a

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large variety of diverse physiological processes, including neurotransmission, blood pressure regulation, inhibition of platelet aggregation, defense mechanisms, smooth muscle relaxation and immune regulation (Bergendi et al., 1999; Patel et al., 1999; Valko et al., 2007). NO $\cdot$ has effects on neuronal transmission and synaptic plasticity in the central nervous system. Nitrosative stress is the state of overproduction of reactive nitrogen species (Klatt et al., 2000; Ridnour et al., 2005). Nitrosative stress may lead to alteration of the protein structure by nitrosylation reactions, and inhibit their normal function, causing cell death (Valko et al., 2007).

Nitric oxide has been implicated as an important mediator of both acute and chronic neurodegenerative injuries (Dawson and Dawson, 1996; Hantraye et al., 1996; Heales et al., 1999). For example, inhibitors of neuronal nitric oxide synthase (nNOS) have been shown to significantly reduce lesions produced in response to focal ischemia (Yoshida et al., 1994) and mice that are deficient in the nNOS gene exhibit significant protection against cerebral ischemia (Huang et al., 1994). And also it contributes to misfolding and accumulating proteins in a number of neurodegenerative disorders including Parkinson's disease, Alzheimer's disease, amyotrophic lateral sclerosis and polyglutamine diseases (Nakamura and Lipton, 2007).

One of the well known NO-releasing drugs is the nitroferricyanide or disodium ditroprusside (SNP) Na₂[NO-Fe(CN)₅], a water-soluble iron nitrosyl complex consisting of a ferrous ion surrounded by five cyanide moieties and a nitrosyl group. It has been used to mainly investigate the apoptotic mechanisms triggered under nitrosative stress conditions and well docu-

mented that NO is the molecule responsible for a lot of pharmacological and toxicological effects derived by SNP (Chen et al., 1991; Friederich and Butterworth, 1995; Li et al., 2004; Cardaci et al., 2008).

In this present study, we investigated whether compounds **1-10** have inhibitory effect against SNP-induced cell death. In this result, the three prenylated flavones (**2**, **3**, and **5**) and prenylated flavanones (**4**, **6**, and **7**) enhanced cell viability in neuroblastoma SH-SY5Y cells against SNP-induced cell death within a non-toxic dose range (Table I).

MATERIALS AND METHODS

General experimental

UV spectra were obtained on a Mecasys Optizen 2120UV. NMR spectra were taken on Varian 500 MHz NMR spectrometer with tetramethylsilane as an internal standard, and chemical shifts are expressed in δ values. Mass spectra were obtained on a Waters Q-TOF micro mass spectrometer. Column chromatography was performed using a silica gel (Kieselgel 60, 70-230 mesh, Merck), RP-18 (12 nm, S-75 μ m, YMC), and thin layer chromatography was performed using a pre-coated silica gel 60 F₂₅₄ (0.25 mm, Merck).

Plant materials

The dry material of Mori Cortex Radicis was purchased from Kyungdong Oriental Herbal Market, Seoul, Korea, on August 2006 and identified by one of us, Dr. Lee, H. J. (Korea Forest Research Institute). A voucher specimen (20060811-MCR) has been deposited at College of Life Sciences and Biotechnology, Korea University, Korea.

Extraction and isolation

The dry material of Mori Cortex Radicis (10 kg) was extracted three times with MeOH at room temperature. Removal of the solvent *in vacuo* yielded an MeOH extract (1.5 kg). The MeOH extract was suspended in distilled water and then partitioned with *n*-hexane and EtOAc (610 g), successively. Column chromatography of the EtOAc-soluble layer (75 g), showing neuroprotective activity (EC₅₀ values, 39.2 mg/mL, SNP-induced), was fractionated over silica gel using a CHCl₃-MeOH mixture with increasing polarity, yielding 12 fractions (SS2-1~12). Fraction SS2-4 was subjected to RP-18 resin eluted with MeCN-H₂O (30:70 to 100:0) to give 14 fractions (SS2-12-1~14). Among the above 14 fractions, compounds **2** (28 mg), **3** (30 mg), and **4** (9 mg) were isolated from fraction SS2-12-9 by column chromatography on Sephadex LH-20 using CHCl₃-MeOH (1:1). Compounds **5** (330 mg) and **6** (65 mg)

Table I. Protective effect of prenylated flavonoids from Mori Cortex Radicis (Moraceae) against SNP-induced cell death in human neuroblastoma SH-SY5Y cells

Compound	EC ₅₀ ^a (μ g/mL)
licoflavone C (1)	>100
cyclomulberrin (2)	4.4 $\pm$ 1.0
neocyclomorusin (3)	5.6 $\pm$ 0.9
sanggenon I (4)	8.0 $\pm$ 1.5
morusin (5)	6.4 $\pm$ 1.9
kuwanon U (6)	8.7 $\pm$ 1.1
kuwanon E (7)	11.9 $\pm$ 1.4
moracin P (8)	>100
moracin O (9)	>100
mulberrofuran Q (10)	>100
carnosine ^b	15.2 $\pm$ 0.6

^aProtective effect was expressed as the mean $\pm$ S.D. of 50% effective concentrations of triplicate determinations

^bPositive control compound

were isolated from fraction SS2-12-10 and SS2-12-12 by column chromatography on silica gel using mixture of CHCl_3 -MeOH as an eluting solvent. Fraction SS2-5 was subjected to flash column chromatography on RP-18 eluting with MeOH- H_2O (40:60 to 90:10) to yield compounds **1** (11 mg) and **7** (70 mg). Fraction SS2-7 was subjected to vacuum liquid chromatography on RP-18 eluting with MeCN- H_2O (20:80 to 100:0) to give nine fractions (SS2-6-1~9). Among the above nine fractions, compounds **8** (5 mg), and **9** (5 mg) were obtained by an additional purification step on the RP-HPLC from fraction SS2-6-3. The HPLC (Varian Prostar 210) system equipped with a PDA detector and an ODS-A (YMC GEL, 250×20 mm, flow rate 8 mL/min) semi-preparative column was used for purification. The mobile phase for HPLC was a mixture of MeOH and H_2O (55:45). Fraction SS2-10 was subjected to flash column chromatography on RP-18 developing with MeCN- H_2O (50:50) to give five fractions (SS2-7-1~5). Compound **10** (21 mg) was isolated from fraction SS2-7-2 by preparative HPLC (62 % MeOH- H_2O).

Licoflavone C (1)

Pale yellow needles; UV (MeOH) λ_{max} (log ϵ) 274 (4.24), 305 (4.12), 326 (4.15); ESIMS m/z 337 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Kajiyama et al., 1992).

Cyclomulberrin (2)

Yellow amorphous powder; UV (MeOH) λ_{max} (log ϵ) 214 (4.79), 274 (4.52), 370 (4.51); ESIMS m/z 419 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Chen et al., 1993).

Neocyclomorusin (3)

Yellow amorphous powder; UV (MeOH) λ_{max} (log ϵ) 217 (4.47), 235 (4.47), 278 (4.49), 335 (4.20); ESIMS m/z 435 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (DMSO- d_6): consistent with published values (Nomura et al., 1976).

Sanggenon I (4)

Amorphous white powder; UV (MeOH) λ_{max} (log ϵ) 215 (4.35), 229 (sh 4.40), 289 (4.22), 318 (sh 3.70); ESIMS m/z 421 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Hano and Nomura, 1983).

Morusin (5)

Yellow crystal; UV (MeOH) λ_{max} (log ϵ) 204 (4.45), 220 (sh 4.39), 279 (4.14), 328 (sh 3.87); ESIMS m/z 419 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (DMSO- d_6): consistent with published values (Tati et al., 2009).

Kuwanon U (6)

Amorphous brown powder; UV (MeOH) λ_{max} (log ϵ) 212 (4.51), 229 (sh 4.25), 289 (4.22), 327 (sh 3.70); ESIMS m/z 437 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Hano et al., 1989).

Kuwanon E (7)

Amorphous brown powder; UV (MeOH) λ_{max} (log ϵ) 213 (4.51), 290 (4.22), 321 (sh 3.70); ESIMS m/z 423 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Nomura and Fukai, 1981).

Moracin P (8)

Brown powder; UV (MeOH) λ_{max} (log ϵ) 219 (3.88), 321 (3.92), 338 (3.89); ESIMS m/z 325 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Hirakura et al., 1986; Dat et al., 2009).

Moracin O (9)

Brown powder; UV (MeOH) λ_{max} (log ϵ) 219 (3.54), 320 (3.59), 335 (3.55); ESIMS m/z 325 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Dat et al., 2009; Hano and Nomura, 1986).

Mulberrofuran Q (10)

Reddish violet powder; UV (MeOH) λ_{max} (log ϵ) 222 (4.41), 280 (4.15), 285 (sh, 4.08), 321 (4.43), 335 (sh 4.29); ESIMS m/z 591 $[\text{M} - \text{H}]^-$; ^1H - and ^{13}C -NMR (acetone- d_6): consistent with published values (Oshima et al., 1981).

Cell culture and treatment

Human neuroblastoma SH-SY5Y cells were obtained from the American Type Culture Collection (ATCC No. CRL-2266). Cells were cultured at 37°C in 5% CO_2 in Dulbecco's modified Eagle's medium (Sigma-Aldrich) supplemented with 10% fetal bovine serum (Gibco-BRL, 100 U/mL penicillin), and 100 $\mu\text{g}/\text{mL}$ antibiotics (Sigma-Aldrich) in a humidified atmosphere with 5% CO_2 and 95% air at 37°C as described previously (Goldberg and Choi, 1993; Fordel et al., 2007). The medium was changed every 3-4 days. Cells were seeded in 96-well culture plates to measure the rate of cell viability at a density 1×10^4 cells/well. And cells were incubated with 2.5 mM SNP with pre-treatment of isolated compounds in this study for 24 h at different concentrations.

Cell viability at different concentration of SNP

Neuroblastoma SH-SY5Y Cells were seeded in 96-well culture plates to measure the rate of cell viability at a density 1×10^4 cells/well. And then cells were

incubated with each final concentration of 1.5 mM, 2 mM, 2.5 mM, 3 mM, and 3.5 mM SNP. After 24 h treatment with SNP, cell viability was measured by MTT assay. From this result, SNP concentration at 2.5 mM (24 h of treatment) was determined for test, showing 56.0% of cell viability (Fig. 2).

Cellular viability measured by MTT assay

At the end of the treatment with 2.5 mM SNP, cell viability (mitochondrial activity) were performed by MTT reduction assay. Neuroblastoma SH-SY5Y cells were incubated for 3 h at 37°C in the presence of MTT (0.5 mg/mL). Active mitochondrial dehydrogenases of living cells cause cleavage and reduction of soluble yellow MTT dye to the insoluble purple formazan, which was extracted in dimethyl sulfoxide (DMSO); the optical density was measured at 540 nm (Miglio et al., 2004).

Statistical analysis

All data are presented as mean $\pm$ S.E.M. Statistical significance was determined by *t*-test (Sigma Plot 10.0 software, SPSS); $p < 0.05$ was considered statistically significant.

RESULTS AND DISCUSSION

The MeOH extract of dried Mori Cortex Radicis was subjected to repeated column chromatographic purifi-

cation and preparative HPLC to afford ten compounds (1-10) of previously known structures. These compounds were identified as licoflavone C (1) (Kajiyama et al., 1992), cyclomulberrin (2) (Chen et al., 1993), neocyclo-morusin (3) (Nomura et al., 1976), sanggenon I (4) (Hano and Nomura, 1983), morusin (5) (Tati et al., 2009), kuwanon U (6) (Hano et al., 1989), kuwanon E

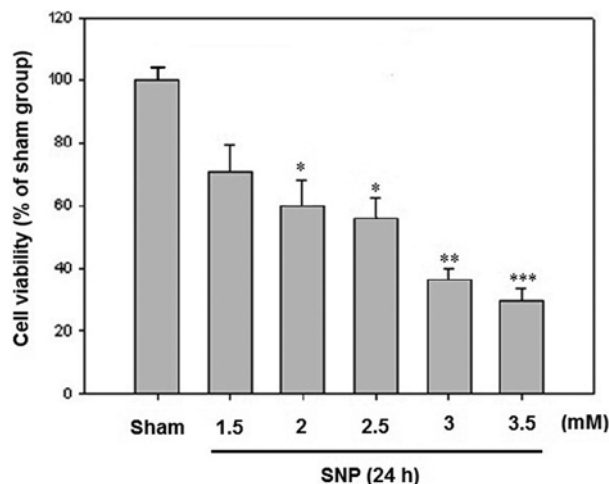


Fig. 2. Cell viability at different concentration of SNP in human neuroblastoma SH-SY5Y cells. Cells were treated with 1.5 mM, 2 mM, 2.5 mM, 3 mM, 3.5 mM SNP for 24 h. And then cell viability was determined by MTT assay. Data represent mean $\pm$ S.E.M. of the three independent experiments. The significance was determined by Student's *t*-test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$ vs sham group).

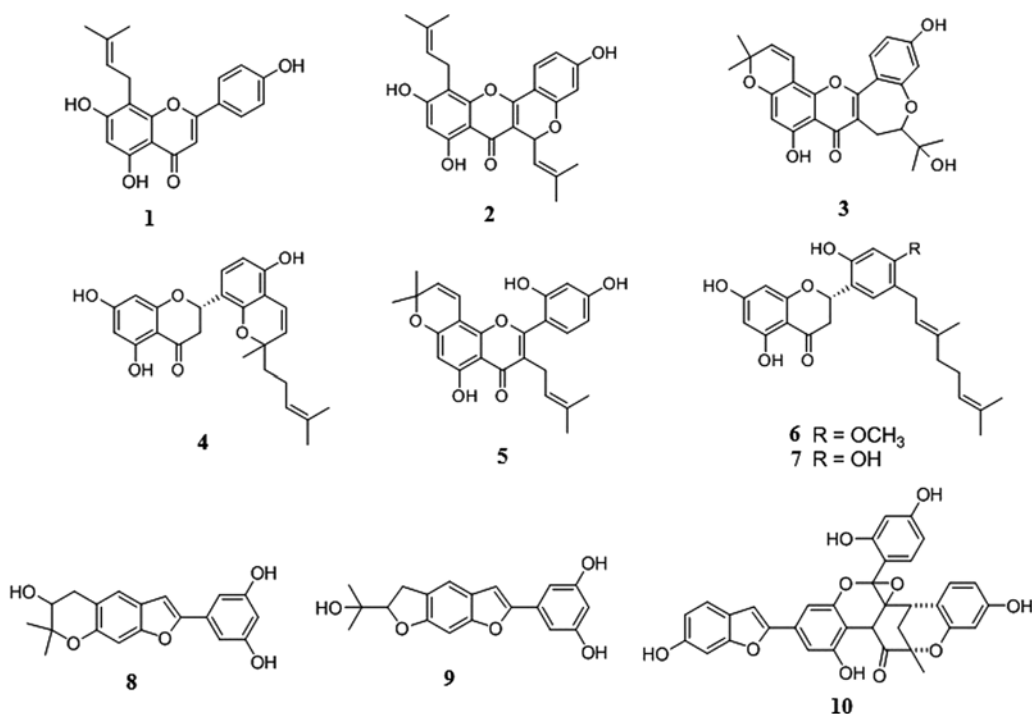


Fig. 1. Structures of compounds 1-10 isolated from Mori Cortex Radicis.

(7) (Nomura and Fukai, 1981), moracin P (8), moracin O (9), and mulberrofuran Q (10) by spectral data interpretation and comparison with literature values (Fig. 1).

In the present study, compounds **1-10** were evaluated for measuring the inhibitory effect against SNP-induced cell death at the concentration from 0.16 to 20 $\mu\text{g/mL}$. Among those compounds, prenylated flavones (**2**, **3**, and **5**) and prenylated flavanones (**4**, **6**, and **7**) enhanced neuronal cell viability from nitrosative stress-induced cell death within a non-toxic dose range.

Previous studies have demonstrated carnosine is significant antioxidant and anti-ischemic properties in the brain, and a scavenger of reactive oxygen species and a protector against neuronal cell death (Stvolinsky and Dobrota, 2000; Dobrota et al., 2005; Stvolinsky et al., 2010) and protects neuronal cells aggregated forms of β -amyloid (Calabrese et al., 2005). In this study, carnosine, a control compound, revealed a protection against SNP-induced neuronal cell death (IC_{50} value of 15.2 $\mu\text{g/mL}$). In this evaluation, protective activities of compounds **2-7** against oxidative stress by SNP

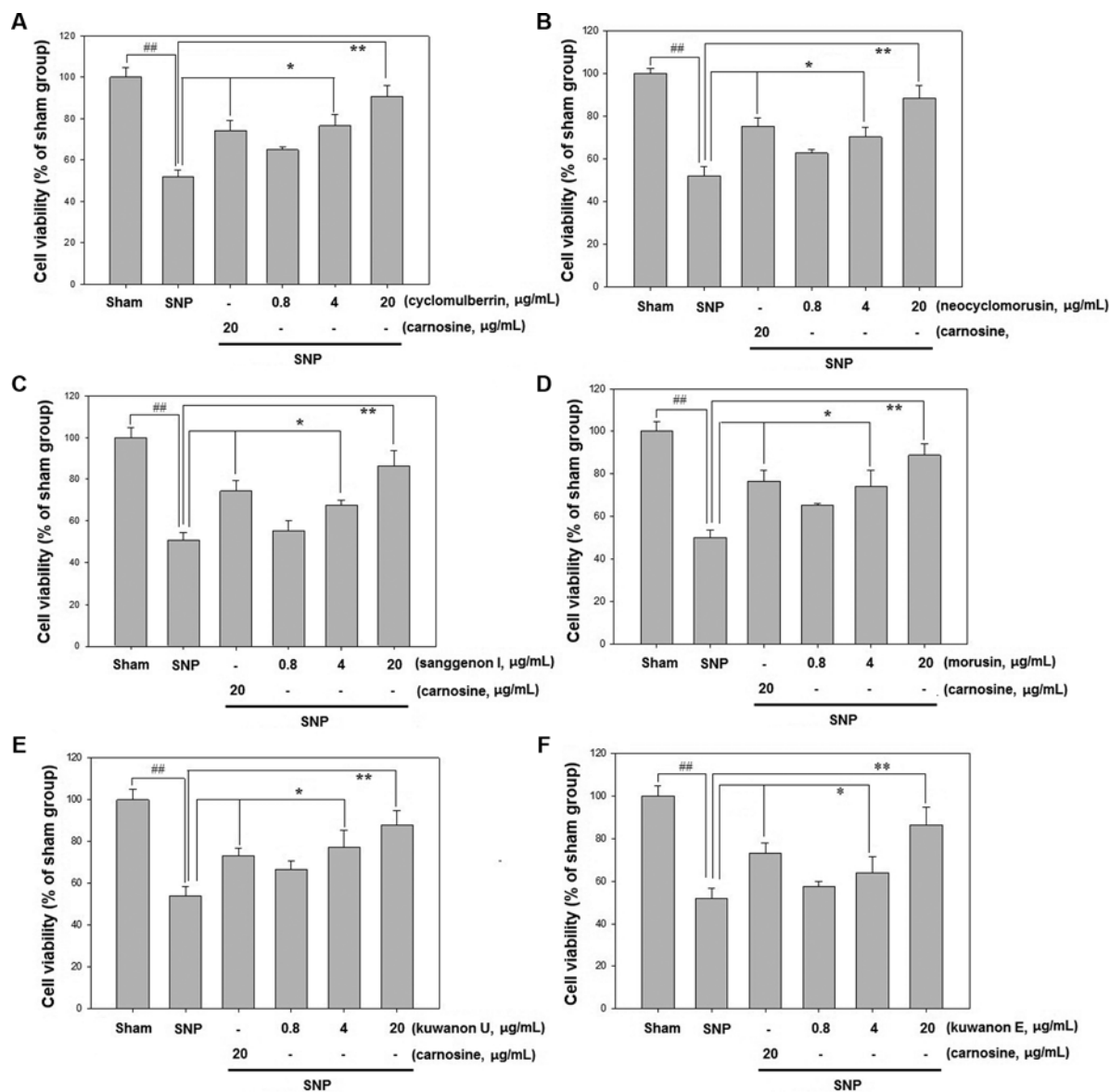


Fig. 3. Inhibition of prenylated flavonoids against SNP-induced human neuroblastoma SH-SY5Y cell death. Among compounds, prenylated flavones (**2**, **3**, and **5**) and prenylated flavanones (**4**, **6**, and **7**) enhanced neuronal cell viability with a non-toxic dose range. Cells were treated with different concentration of compounds **2-7** and exposed to SNP (2.5 mM of treatment) for 24 h. Cell viability was determined by MTT assay. Data represent mean $\pm$ S.E.M. of the three independent experiments. The significance was determined by Student's *t*-test (##*p* < 0.01 vs sham group; **p* < 0.05, ***p* < 0.01 vs SNP-only treated group).

were more effective than carnosine, a positive control compound (Fig. 3).

It is well reported that NO is the molecule responsible for a lot of pharmacological and toxicological effects elicited by SNP (Cardaci et al., 2008). NO and its metabolites were found to induce apoptotic cell death both in neuronal cell lines and in primary neurons (Zhang and Zhao, 2003). Also, NO is a main factor contributing to damages in neuronal sites in ischemic stroke, demyelinating diseases and other neurodegenerative disorders (Lim et al., 2009).

In our search for neuroprotective agents of natural origin, a MeOH extract of Mori Cortex Radicis significantly increased the cell viability of SH-SY5Y human neuroblastoma cells exposed to SNP (EC_{50} 39.2 μ g/mL). Bioactivity-guided fractionation of the MeOH extract of Mori Cortex Radicis using this assay led to the isolation of ten known compounds (1-10).

Among the tested compounds, C-3 prenylated flavones, cyclomulberrin (2), neocyclomorusin (3), and morusin (5), improved cell viability in dose-dependent manner against SNP-induced cell death in SH-SY5Y cells, with an EC_{50} values of 4.4, 5.6, and 6.4 μ g/mL, respectively. Also, prenylated flavanones, sanggenon I (4), kuwanons U (6), and E (7), exhibited inhibitory activity, with an EC_{50} values ranging 7.9 to 12.0 μ g/mL. However, licoflavone (1, which lacks the prenyl group at C-3) and three 2-arylbenzofurans (8-10) did not have an inhibitory effect on NO production even at 100 μ g/mL. The order of enhanced cell viability in our study was as follow: C-3 prenylated flavones (2, 3, and 5) > prenylated flavanones (4, 6, and 7) > 2-arylbenzofurans (8-10) and flavone (1).

Although further extensive studies regarding the effects of these compounds from Mori Cortex Radicis on nitrosative stress-mediated diseases are necessary, we expect some prenylated flavonoids from Mori Cortex Radicis might protect cells against nitric oxide-induced oxidative stress-associated injuries which lead to severe neuronal damages.

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